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Sequential combustion gas turbine engine including a steam system for injecting steam to each combustor

Abstract

A turbine engine includes a compressor section for providing a compressed air flow, a fuel system for providing a primary combustor fuel supply, and a secondary combustor fuel supply. A primary combustor, located downstream of the compressor section, combusts the compressed air flow and the primary combustor fuel supply to generate primary combustion products. A turbine section, located downstream of the primary combustor, includes a turbine and a secondary combustor that further combusts the primary combustion products and the fuel from the secondary combustor fuel supply. A steam system provides a steam supply to the secondary combustor. The secondary combustor further combusts the primary combustion products and the secondary combustor fuel supply to generate secondary combustion products, and the turbine section is rotated by the secondary combustor steam supply and the inter-turbine combustion products.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4081956	12/1977	Baker et al.	N/A	N/A
4272953	12/1980	Rice	60/774	F01K 23/103
4369624	12/1982	Hamm et al.	N/A	N/A
4571935	12/1985	Rice	60/39.182	F02C 6/18
5054279	12/1990	Hines	60/39.5	F01K 21/047
5329758	12/1993	Urbach	60/39.55	F01K 21/047
5775091	12/1997	Bannister	60/39.465	F01D 5/084
5896740	12/1998	Shouman	N/A	N/A
5906095	12/1998	Frutschi et al.	N/A	N/A
6089024	12/1999	Hatanaka	N/A	N/A
7726133	12/2009	Hannemann et al.	N/A	N/A
8613186	12/2012	Koganezawa et al.	N/A	N/A
11603798	12/2022	Terwilliger et al.	N/A	N/A
2002/0148213	12/2001	Yu	60/39.17	F02C 3/365
2006/0207262	12/2005	Firey	N/A	N/A
2007/0003897	12/2006	Koizumi et al.	N/A	N/A
2007/0033918	12/2006	Benz	60/39.12	F01K 23/10
2007/0039468	12/2006	Benz	95/116	F02C 6/10
2007/0271898	12/2006	Little	60/39.182	F02C 6/18
2009/0260368	12/2008	Benz	60/780	F25J 3/04575
2014/0137564	12/2013	Scipio	60/39.53	F02C 3/30
2016/0115826	12/2015	Eroglu	N/A	N/A
2018/0252410	12/2017	Zelina et al.	N/A	N/A
2018/0266333	12/2017	Gassner	N/A	F02C 9/20

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4407161	12/2023	EP	N/A
2236145	12/1990	GB	N/A
2022090990	12/2021	WO	N/A

OTHER PUBLICATIONS

Zelina et al., "Exploration of Compact Combustors for Reheat Cycle Aero Engine Applications," Proceedings of ASME Turbo Expo '06: 51st ASME International Gas Turbine and Aeroengine Congress and Exposition, p. 1-11 (2006). cited by applicant

Zelina, "Technical Article: Revolutionary Combustors for Future Propulsion Systems," Global Gas Turbine News, vol. 53, No. 4, p. 51 (2023). cited by applicant

Daggett, "Water Injection Feasibility for Boeing 747 Aircraft," National Aeronautics and Space Administration, p. i-vi and 1-98 (2005). cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to a turbine engine including a steam system.

BACKGROUND

(2) Turbine engines generally include a fan and a core section arranged in flow communication with one another. A combustor is arranged in the core section to generate combustion gases for driving a turbine in the core section of the turbine engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other features and advantages will be apparent from the following, more particular, description of various exemplary embodiments, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(2) FIG. 1 is a schematic cross-sectional view of a turbine engine with a steam system, taken along a longitudinal centerline axis of the turbine engine, according to an embodiment of the present disclosure.

(3) FIG. 2 is a schematic view of a portion of a turbine engine, including a steam system, a fuel system, and a control system, according to an embodiment of the present disclosure.

(4) FIG. 3 is a schematic view of a portion of a turbine engine, including a steam system, a fuel system, and a control system, according to an embodiment of the present disclosure.

(5) FIG. 4 is a schematic view of a portion of a turbine engine, including a steam system, a fuel system, and a control system, according to an embodiment of the present disclosure.

(6) FIG. 5 is a schematic cross-sectional view of a turbine section of a turbine engine, taken along a longitudinal centerline axis of the turbine engine, the turbine section including a steam system and a fuel system, according to an embodiment of the present disclosure.

(7) FIG. 6 is a schematic view of a tangential radial inflow combustor, according to an embodiment of the present disclosure.

(8) FIG. 7 is a schematic view of a trapped vortex cavity combustor, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(9) Features, advantages, and embodiments of the present disclosure are set forth or apparent from

a consideration of the following detailed description, drawings, and claims. Moreover, the following detailed description is exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.

(10) Various embodiments are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the spirit and scope of the present disclosure.

(11) As used herein, the terms “first,” “second,” “third,” and “fourth” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

(12) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

(13) The terms “forward” and “aft” refer to relative positions within a turbine engine or a vehicle, and refer to the normal operational attitude of the turbine engine or vehicle. For example, with regard to a turbine engine, forward refers to a position closer to an engine inlet, and aft refers to a position closer to an engine nozzle or an exhaust.

(14) The terms “coupled,” “fixed,” “attached,” and the like, refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

(15) As used herein, the terms “low,” “mid” (or “mid-level”), and “high,” or their respective comparative degrees (e.g., “lower” and “higher,” where applicable), when used with compressor, turbine, shaft, fan, or turbine engine components, each refers to relative pressures, relative speeds, relative temperatures, and/or relative power outputs within an engine unless otherwise specified. For example, a “low-power” setting defines the engine configured to operate at a power output lower than a “high-power” setting of the engine, and a “mid-level power” setting defines the engine configured to operate at a power output higher than a “low-power” setting and lower than a “high-power” setting. The terms “low,” “mid” (or “mid-level”) or “high” in such aforementioned terms may additionally or alternatively, be understood as relative to minimum allowable speeds, pressures, or temperatures, or minimum or maximum allowable speeds, pressures, or temperatures relative to normal, desired, steady state, etc., operation of the engine.

(16) The various power levels of the turbine engine detailed herein are defined as a percentage of a sea level static (SLS) maximum engine rated thrust. Low-power operation includes, for example, less than thirty percent (30%) of the SLS maximum engine rated thrust of the turbine engine. Mid-level power operation includes, for example, thirty percent (30%) to eighty-five percent (85%) of the SLS maximum engine rated thrust of the turbine engine. High-power operation includes, for example, greater than eighty-five percent (85%) of the SLS maximum engine rated thrust of the turbine engine. The values of the thrust for each of the low-power operation, the mid-level power operation, and the high-power operation of the turbine engine are exemplary only, and other values of the thrust can be used to define the low-power operation, the mid-level power operation, and the high-power operation.

(17) Here and throughout the specification and claims, range limitations are combined, and interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(18) The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(19) As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the turbine engine. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the

centerline of the turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the turbine engine.

(20) The term “combustion products” refers to all products exiting a combustor in a turbine engine, such products include all outflowing products, such as, but not limited to, the gas products from the chemical reaction of combustion such as $O_{2.0}$, $N_{2.0}$, $CO_{2.0}$, CO, NO_x , water vapor, soot, unspent fuel in a fuel-rich burn condition, or yet to be combusted air in a fuel-lean burn condition

(21) A high bypass ratio and high overall pressure ratios are incorporated into turbine engines in order to minimize fuel consumption and $CO_{2.0}$ emissions. Consequently, turbine engines have resulting higher operational pressures and temperatures, which, in turn, generate greater pollutant emissions, including nitrogen oxide (NO_x) emissions. Additional emissions of turbine engines as applied in aircraft include water vapor, which, upon exiting the turbine engine, condenses, forming condensation trails, also known as contrails. Contrail generation may be undesirable.

(22) Accordingly, the present disclosure provides a turbine engine having one or more secondary combustors for introducing fuel and steam thereto. The one or more secondary combustors may employ a reheat cycle to reduce the overall maximum temperature of the turbine engine, thereby decreasing the emission of NO_x and other pollutants. The one or more secondary combustors of the present disclosure may include an inter-turbine combustor, an inter-stage combustor, or both an inter-turbine combustor and an inter-stage combustor.

(23) Injecting steam at the one or more secondary combustors may lower the operating temperature of the turbine engine, further reducing pollutant emissions. Additionally, injecting steam increases mass flow rate, thereby increasing thrust. Beyond reducing pollutants, lower engine temperatures may contribute to enhanced engine durability by reducing damage caused by high temperature gases flowing past various engine components, including, for example, the combustor liners and the high-pressure turbine, and the low-pressure turbine. Injecting steam at the one or more secondary combustors may allow reducing steam addition in the main combustor to improve combustion flameout or blowout. Blowout is defined as a state in which the combustor can no longer sustain combustion. Steam may be introduced into the one or more secondary combustors at any ratio from zero (no steam), up the point of blowout. A preferable steam ratio in the one or more secondary combustors may be 0.1 to 0.5 of steam to fuel ratio.

(24) The present disclosure also includes injecting fuel in the one or more secondary combustors. In the event of loss of steam to the one or more secondary combustors, injecting fuel in the primary combustor or the one or more secondary combustors, or both, may assist in maintaining engine operation by varying the amount of fuel and, thus, varying heat addition, to the engine. In some examples, steam is injected in an amount from 0.1 to 0.5 of steam to fuel ratio.

(25) Referring now to the drawings, FIG. 1 is a schematic cross-sectional view of a turbine engine 10 including a steam system 92, taken along a longitudinal centerline axis 12 (provided for reference) of the turbine engine 10, according to an embodiment of the present disclosure. As shown in FIG. 1, the turbine engine 10 has an axial direction A (extending parallel to the longitudinal centerline axis 12) and a radial direction R that is normal to the axial direction A. In general, the turbine engine 10 includes a fan section 14 and a core turbine engine 16 disposed downstream from the fan section 14.

(26) The core turbine engine 16 includes an outer casing 18 that is substantially tubular and defines an annular core inlet 20. As schematically shown in FIG. 1, the outer casing 18 encases, in serial flow relationship, a compressor section 21 including a booster or a low-pressure compressor (LPC) 22 followed downstream by a high-pressure compressor (HPC) 24, a primary combustor 26, a turbine section 27, including a high-pressure turbine (HPT) 28, followed downstream by a low-pressure turbine (LPT) 30, and one or more core exhaust nozzles 32. A high-pressure (HP) shaft 34 or a spool drivingly connects the HPT 28 to the HPC 24 to rotate the HPT 28 and the HPC 24 in

unison. The HPT **28** is drivingly coupled to the HP shaft **34** to rotate the HP shaft **34** when the HPT **28** rotates. A low-pressure (LP) shaft **36** drivingly connects the LPT **30** to the LPC **22** to rotate the LPT **30** and the LPC **22** in unison. The LPT **30** is drivingly coupled to the LP shaft **36** to rotate the LP shaft **36** when the LPT **30** rotates. The compressor section **21**, the primary combustor **26**, the turbine section **27**, and the one or more core exhaust nozzles **32** together define a core air flow path **33**.

(27) For the embodiment depicted in FIG. **1**, the fan section **14** includes a fan **38** (e.g., a variable pitch fan) having a plurality of fan blades **40** coupled to a disk **42** in a spaced apart manner. As depicted in FIG. **1**, the fan blades **40** extend outwardly from the disk **42** generally along the radial direction R. Each fan blade **40** is rotatable relative to the disk **42** about a pitch axis P by virtue of the fan blades **40** being operatively coupled to an actuator **44** configured to collectively vary the pitch of the fan blades **40** in unison. The fan blades **40**, the disk **42**, and the actuator **44** are together rotatable about the longitudinal centerline axis **12** via a fan shaft **45** that is powered by the LP shaft **36** across a power gearbox, also referred to as a gearbox assembly **46**. The gearbox assembly **46** is shown schematically in FIG. **1**. The gearbox assembly **46** includes a plurality of gears for adjusting the rotational speed of the fan shaft **45** and, thus, the fan **38** relative to the LP shaft **36**.

(28) Referring still to the exemplary embodiment of FIG. **1**, the disk **42** is covered by a rotatable fan hub **48** aerodynamically contoured to promote an airflow through the plurality of fan blades **40**. In addition, the fan section **14** includes an annular fan casing or a nacelle **50** that circumferentially surrounds the fan **38**, at least a portion of the core turbine engine **16**, or both. The nacelle **50** is supported relative to the core turbine engine **16** by a plurality of circumferentially spaced outlet guide vanes **52**. Moreover, a downstream section **54** of the nacelle **50** extends over an outer portion of the core turbine engine **16** to define a bypass airflow passage **56** therebetween. The one or more core exhaust nozzles **32** may extend through the nacelle **50** and be formed therein. In this example, the one or more core exhaust nozzles **32** include one or more discrete nozzles that are spaced circumferentially about the nacelle **50**. Other arrangements of the one or more core exhaust nozzles **32** may be used including, for example, a single core exhaust nozzle that is annular, or partially annular, about the nacelle **50**.

(29) During operation of the turbine engine **10**, a volume of air **58** enters the turbine engine **10** through an inlet **60** of the nacelle **50** or the fan section **14**. As the volume of air **58** passes across the fan blades **40**, a first portion of air **62**, also referred to as bypass air **62**, is directed or routed into the bypass airflow passage **56**, and a second portion of core air **64**, also referred to as core air **64**, is directed or is routed into the upstream section of the core air flow path **33**, or, more specifically, into the annular core inlet **20**. The ratio between the first portion of air **62** and the second portion of core air **64** is known as a bypass ratio. In some embodiments, the bypass ratio may be greater than 18:1, enabled by the steam system **92**, as detailed further below. The pressure of the core air **64** is increased by the LPC **22**, generating compressed air **65**, and the compressed air **65** is routed through the HPC **24** and further compressed before being directed into the primary combustor **26**, where the compressed air **65** is mixed with fuel **67** and burned to generate primary combustion products **66**, also referred to as primary combustion gases **66**.

(30) The LPC **22**, the HPC **24**, or both the LPC **22** and the HPC **24** may include one or more stages, with each subsequent stage further compressing the compressed air **65**. The HPC **24** may have a compression ratio greater than 20:1, preferably, in a range of 20:1 to 40:1. The compression ratio is a ratio of a pressure of a last stage of the HPC **24** to a pressure of a first stage of the HPC **24**. The compression ratio greater than 20:1 is enabled by the steam system **92**, as detailed further below.

(31) The primary combustion products **66** are routed into the HPT **28** and expanded through the HPT **28** where a portion of thermal energy or kinetic energy from the primary combustion products **66** is extracted via sequential stages of HPT stator vanes **68** that are coupled to the outer casing **18** and to a plurality of HPT rotor blades **70** that are coupled to the HP shaft **34**, thus, causing the HP shaft **34** to rotate, thereby supporting operation of the HPC **24**. The primary combustion products

66 are then routed into the LPT **30** and expanded through the LPT **30**. Here, a second portion of thermal energy or kinetic energy is extracted from the primary combustion products **66** via sequential stages of LPT stator vanes **72** that are coupled to the outer casing **18** and to a plurality of LPT rotor blades **74** that are coupled to the LP shaft **36**, thus, causing the LP shaft **36** to rotate, thereby supporting operation of the LPC **22** and rotation of the fan **38** via the gearbox assembly **46**. The HPT **28**, the LPT **30**, or both the HPT **28** and the LPT **30** may have one or more stages. The HPC **24** having a compression ratio in a range of 20:1 to 40:1 results in the HPT **28** having a pressure expansion ratio in a range of 1.5:1 to 4:1 and the LPT **30** having a pressure expansion ratio in a range of 4.5:1 to 28:1.

(32) The primary combustion products **66** are subsequently routed through the one or more core exhaust nozzles **32** of the core turbine engine **16** to provide propulsive thrust. Simultaneously with the flow of the core air **64** through the core air flow path **33**, the bypass air **62** is routed through the bypass airflow passage **56** before being exhausted from a fan bypass nozzle **76** of the turbine engine **10**, also providing propulsive thrust. The primary combustor **26**, the HPT **28**, the LPT **30**, and the one or more core exhaust nozzles **32** at least partially define a hot gas path **78** for routing the primary combustion products **66** through the core turbine engine **16**.

(33) As noted above, the compressed air **65** (i.e., the core air **64**) is mixed with the fuel **67** in the primary combustor **26** to generate a fuel and air mixture, and combusted, generating primary combustion products **66** (i.e., combustion products). The fuel **67** can include any type of fuel used for turbine engines, such as, for example, sustainable aviation fuels (SAF) including biofuels, Jet A, or other hydrocarbon fuels. The fuel **67** also may be a hydrogen-based fuel (H.sub.2), that may include blends with hydrocarbon fuels. In some embodiments, the hydrogen fuel may comprise substantially pure hydrogen molecules (i.e., diatomic hydrogen). The fuel **67** may be a cryogenic fuel. For example, in examples with hydrogen fuel, the hydrogen fuel may be stored in a liquid phase at cryogenic temperatures.

(34) The turbine engine **10** includes a fuel system **80** for providing the fuel **67** to the primary combustor **26**. The fuel system **80** includes a fuel tank **82** for storing the fuel **67** therein, a fuel manifold **90**, and a primary combustor fuel supply **84**. The fuel tank **82** can be located on an aircraft (not shown) to which the turbine engine **10** is attached. Although a single fuel tank **82** is shown in FIG. 1, the fuel system **80** can include any number of fuel tanks **82**. The fuel manifold **90** apportions the flow of fuel **67** from the fuel tank **82** for the primary combustor **26**. The primary combustor fuel supply **84** provides fuel **67** from the fuel manifold **90** to the primary combustor **26**. The primary combustor fuel supply **84** includes one or more lines, conduits, pipes, tubes, etc., configured to carry the fuel **67** from the fuel tank **82** to the primary combustor **26**. The primary combustor fuel supply **84** also includes a pump **86** to induce the flow of the fuel **67** through the primary combustor fuel supply **84** to the primary combustor **26**. In this way, the pump **86** causes the fuel **67** to flow from the fuel tank **82**, through the primary combustor fuel supply **84**, and into the primary combustor **26**. Alternatively, the pump **86** may be integrated into the fuel manifold **90**. The fuel system **80** and, more specifically, the fuel tank **82** and the primary combustor fuel supply **84**, either collectively or individually, may be a fuel source for the primary combustor **26**.

(35) In some embodiments, for example, when the fuel **67** is a hydrogen fuel, the fuel system **80** includes one or more vaporizers **88** (illustrated as optionally included by dashed lines) and a fuel manifold **90** in fluid communication with the primary combustor fuel supply **84**. In such an example, the hydrogen fuel is stored in the fuel tank **82** as liquid hydrogen fuel. The one or more vaporizers **88** heat the liquid hydrogen fuel. The one or more vaporizers **88** may be positioned in the flow path of the fuel **67** between the fuel tank **82** and the fuel manifold **90**, and may be located downstream of the pump **86**. Alternatively, the vaporizers may be incorporated into the structure of the fuel manifold **90**. The one or more vaporizers **88** are in thermal communication with at least one heat source, such as, for example, waste heat from the turbine engine **10** and/or from one or more systems of the aircraft (not shown). The one or more vaporizers **88** heat the liquid hydrogen

fuel and the liquid hydrogen fuel is converted into a gaseous hydrogen fuel within the one or more vaporizers **88**. The primary combustor fuel supply **84** directs the gaseous hydrogen fuel into the primary combustor **26**.

(36) In some embodiments, for example, the fuel manifold **90** is positioned downstream of the one or more vaporizers **88** and the pump **86**. The fuel manifold **90** receives hydrogen fuel in a substantially completely gaseous phase, or in a substantially completely supercritical phase. The fuel manifold **90** provides the flow of fuel to the primary combustor **26** in a desired manner. More specifically, the fuel manifold **90** provides a desired volume of hydrogen fuel at, for example, a desired flow rate, to one or more fuel injectors that inject the hydrogen fuel into the primary combustor **26**. The fuel system **80** can include additional components for supplying the fuel **67** from the fuel tank **82** to the primary combustor **26**, such as, for example, additional pumps, valves, fuel lines, and the like.

(37) The turbine engine **10** includes the steam system **92**. The steam system **92** provides steam to components of the turbine engine **10** in order to improve or to optimize one or more parameters of engine performance.

(38) Although not illustrated in FIG. **1**, the turbine engine **10** may include a control system, such as described in more detail to follow.

(39) The turbine engine **10** depicted in FIG. **1** is by way of example only. In other exemplary embodiments, the turbine engine **10** may have any other suitable configuration. For example, in other exemplary embodiments, the fan **38** may be configured in any other suitable manner (e.g., as a fixed pitch fan) and further may be supported using any other suitable fan frame configuration. Moreover, in other exemplary embodiments, any other suitable number or configuration of compressors, turbines, shafts, or a combination thereof may be provided. In still other exemplary embodiments, aspects of the present disclosure may be incorporated into any other suitable turbine engine, such as, for example, turbofan engines, propfan engines, and/or turboprop engines.

(40) FIG. **2** illustrates a schematic partial view of a turbine engine **100**. The turbine engine **100** is substantially similar to the turbine engine **10** described with respect to FIG. **1**, with the addition of a secondary combustor **194**, also referred to herein as an inter-turbine combustor **194**. Accordingly, the same reference numerals will be used for components of the turbine engine **100** that are the same as or similar to the components of the turbine engine **10** discussed above. The description of these components above also applies to this embodiment, and a detailed description of these components is omitted here.

(41) The turbine engine **100** includes the secondary combustor **194**, a steam system **200**, a fuel system **300**, and a control system **400**. In the turbine engine **100** of FIG. **2**, the secondary combustor **194** is an inter-turbine combustor **194**, also referred to as an inter-turbine burner (ITB) **194**.

(42) To support combustion in the turbine engine **100**, the fuel system **300** includes a fuel supply **302**, a fuel manifold **304**, a primary combustor fuel supply **306**, and a secondary combustor fuel supply **308**. The secondary combustor fuel supply **308** of FIG. **2** is an inter-turbine combustor fuel supply **308**. The primary combustor fuel supply **306** provides the fuel from the fuel supply **302** to the primary combustor **26** and the inter-turbine combustor fuel supply **308** provides the fuel from the fuel supply **302** to the inter-turbine combustor **194**. The fuel manifold **304** apportions the flow of fuel from the fuel supply **302** between the primary combustor fuel supply **306** and the inter-turbine combustor fuel supply **308**. The fuel supply **302**, the fuel manifold **304**, or both may include previously discussed components such as a fuel tank, a pump, a vaporizer, or combinations thereof.

(43) The steam system **200** includes a steam supply **202**, a steam manifold **204**, a primary combustor steam supply **206**, and a secondary combustor steam supply **208**. The secondary combustor steam supply **208** of FIG. **2** is an inter-turbine combustor steam supply **208**. The primary combustor steam supply **206** provides the steam from the steam supply **202** to the primary

combustor **26** and the inter-turbine combustor steam supply **208** provides the steam from the steam supply **202** to the inter-turbine combustor **194**. The steam manifold **204** apportions the flow of steam received from the steam supply **202** between the primary combustor steam supply **206** and the inter-turbine combustor steam supply **208**.

(44) The steam supply **202** provides steam for the steam system **200** from one or more sources, such as, but not limited to, a condensing waste heat recovery system, an onboard water supply, an onboard steam supply, or other suitable supply of liquid water, gaseous water, or combinations thereof. The steam supply **202** provides steam to the steam manifold **204**, which apportions the flow of steam between the primary combustor **26** and the inter-turbine combustor **194** through the primary combustor steam supply **206** and the inter-turbine combustor steam supply **208**, respectively.

(45) In some examples, a turbine engine with the primary combustor **26** and the inter-turbine combustor **194** may, alternatively, have an inter-turbine combustor steam supply **208** and no primary combustor steam supply **206**. In such case, the inter-turbine combustor steam supply **208** provides the benefits of increasing energy at the inter-turbine combustor steam supply **208** and downstream of the inter-turbine combustor steam supply **208**, and reducing temperature at or near the exit of the inter-turbine combustor **194**.

(46) As described previously, air is progressively compressed through the LPC **22** and the HPC **24** of the turbine engine **100** and then flows to the primary combustor **26** where the air is mixed with the fuel from the primary combustor fuel supply **306** and the steam from the primary combustor steam supply **206**. The mixture of the steam, the fuel, and the air is combusted, producing the primary combustion products **66**. The primary combustion products **66** flow into the HPT **28**, wherein the HPT **28** extracts thermal or kinetic energy from the primary combustion products **66**, supporting the rotation of the HP shaft **34** and the HPC **24**.

(47) At the exit of the HPT **28**, the primary combustion products **66** have a lower temperature relative to the temperature at the exit of the primary combustor **26**. Following the HPT **28**, the primary combustion products **66** flow into the inter-turbine combustor **194**. In the inter-turbine combustor **194**, the primary combustion products **66** mix with the fuel from the inter-turbine combustor fuel supply **308** and the steam from the inter-turbine combustor steam supply **208**. The mixture of the primary combustion products **66**, the steam, and the fuel has a lower temperature than the primary combustion products **66** alone. The mixture is combusted in the inter-turbine combustor **194** generating secondary combustion products **167**, also referred to herein as inter-turbine combustion products **167**. The inter-turbine combustion products **167** flow into the LPT **30**, where the LPT **30** extracts a portion of the thermal energy, the kinetic energy, or both the thermal energy and the kinetic energy from the inter-turbine combustion products **167**. The flow of the inter-turbine combustion products **167** through the LPT **30** contributes to the rotation of the LPT **30**, the LPC **22** and the LP shaft **36**. Following the LPT **30**, the inter-turbine combustion products **167** resume the flow path as described with respect to FIG. **1**.

(48) The inter-turbine combustor **194** provides a secondary combustion process in addition to the primary combustion process of the primary combustor **26**. Such a secondary combustion process is also referred to as a reheat cycle. By conducting a portion of the total combustion in each of the primary combustor **26** and the inter-turbine combustor **194**, the total combustion is separated into multiple separate combustion processes with an intermediate temperature reduction in the primary combustion products **66** occurring through the energy extraction in the first turbine, in this case, the HPT **28**, disposed between the primary combustor **26** and the inter-turbine combustor **194**.

(49) This separated combustion may reduce the maximum temperature at the primary combustor **26** relative to a substantially similar capacity turbine engine without a secondary combustor (such as the inter-turbine combustor **194**). A lower maximum temperature may serve to improve durability of the turbine engine **100**, extend the life of the turbine engine **100**, extend the life of the components of the turbine engine **100** susceptible to high heat, or any combination thereof.

Examples of the components susceptible to high heat may include the surfaces and the components of the primary combustor **26**, the surfaces and the components of the HPT **28**, or both. The surfaces may be, for example, but not limited to, inner surfaces, inner liners, surface coatings, or the like. The components may include, for example, but not limited to, bearings. Additionally or alternatively, the lower maximum temperature of the primary combustor may serve to reduce pollutants, such as nitrogen oxides.

(50) The maximum temperature of the inter-turbine combustion products **167** occurs at or near the exit of the inter-turbine combustor **194**. As mentioned, the HPT **28** lowers the temperature of the primary combustion products **66** prior to combustion in the inter-turbine combustor **194**. This results in lowering the overall maximum temperature of the turbine engine **100**, including lowering the temperature of the flow therethrough, which may occur either at or near the exit of the primary combustor **26**, at or near the exit of the inter-turbine combustor **194**, or both, relative to a substantially similar capacity turbine engine without a secondary combustor (such as the inter-turbine combustor **194**).

(51) The control system **400** of FIG. **2** includes a controller **402** that may be communicatively and operatively coupled to the fuel system **300**, the steam system **200**, the primary combustor **26**, and the secondary combustor **194**. For example, the controller **402** may send, receive, or both send and receive a primary combustor data signal **406** from the primary combustor **26**, a secondary combustor data signal **408** (also referred to herein as an inter-turbine combustor data signal **408**) from the secondary combustor **194**, a fuel manifold control signal **412** to the fuel system **300**, and a steam manifold control signal **414** to the steam system **200**. The controller **402** relies on the primary combustor data signal **406** and the inter-turbine combustor data signal **408** to determine the appropriate fuel manifold control signal **412** and to determine the appropriate steam manifold control signal **414**. The primary combustor data signal **406** and the inter-turbine combustor data signal **408** may provide operational state information about the primary combustor **26** and the inter-turbine combustor **194**, respectively.

(52) Such signals may include information related to a temperature, a pressure, a mass flow rate, a moisture content, NO.sub.x emissions, measurable values relevant to the performance of the turbine engine **100** or relevant to the state of the turbine engine at a location of measurement within the turbine engine **100**, or combinations thereof. Although not illustrated, the signals may reflect data collected from other locations in the turbine engine **100**, such as, for example, immediately aft of the primary combustor **26** or immediately aft of the HPT **28**, to provide indirect measuring of the state or the condition of the primary combustor **26** or the secondary combustor **194**. The primary combustor data signal **406**, for example, may include temperature data taken within the primary combustor **26**. The temperature information may allow for control of the fuel system **300**, the steam system **200**, the combustors, or combinations thereof to ensure the temperature is maintained within a predetermined range or below a high temperature threshold. The controller **402** may process the signals received from the primary combustor data signal **406**, the inter-turbine combustor data signal **408**, or both, to determine the appropriate fuel manifold control signal **412** and the steam manifold control signal **414** to send to the fuel manifold **304** and the steam manifold **204**, respectively. The appropriate fuel manifold control signal **412** and the appropriate steam manifold control signal **414** may depend on the operating condition of the turbine engine **100** and the information gathered (e.g., the data signals) from the combustors, as described in more detail to follow.

(53) The controller **402** may be an engine controller. The controller **402** is configured to operate various aspects of the turbine engine **100**, including, in the embodiments discussed herein, the fuel system **300**, the steam system **200**, the primary combustor **26**, and the secondary combustor **194**. The controller **402** may be a Full Authority Digital Engine Control (FADEC). In this embodiment, the controller **402** is a computing device having one or more processors **404** and one or more memories **405**. The processor **404** may be any suitable processing device, including, but not limited

to, a microprocessor, a microcontroller, an integrated circuit, a logic device, a programmable logic controller (PLC), an application-specific integrated circuit (ASIC), and/or a Field Programmable Gate Array (FPGA). The memory **405** may include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, a computer-readable non-volatile medium (e.g., a flash memory), a RAM, a ROM, hard drives, flash drives, and/or other memory devices.

(54) The memory **405** may store information accessible by the processor **404**, including computer-readable instructions that may be executed by the processor **404**. The instructions may be any set of instructions or a sequence of instructions that, when executed by the processor **404**, causes the processor **404** and the controller **402** to perform operations. In some embodiments, the instructions may be executed by the processor **404** to cause the processor **404** to complete any of the operations and functions for which the controller **402** is configured, as described herein. The instructions may be software written in any suitable programming language or may be implemented in hardware. Additionally or alternatively, the instructions may be executed in logically and/or virtually separate threads on the processor **404**. The memory **405** may further store data that may be accessed by the processor **404**. The memory **405** may apply certain rules or control logic, which control the apportionment of fuel and steam. Such rules may apply to engine power, emissions targets, engine safety parameters, engine efficiency, or any other such relevant measurable output.

(55) The technology discussed herein makes reference to computer-based systems and actions taken by, and information sent to and from, computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between components and among components. For instance, processes discussed herein may be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications may be implemented on a single system or distributed across multiple systems. Distributed components may operate sequentially or in parallel.

(56) The control system **400** allows for control of the turbine engine **100** during different power requirements and operating requirements at different phases of operation. For example, when included in an aircraft, different phases of flight have different thrust requirements for the turbine engine **100**. Accordingly, the primary combustor **26** and the inter-turbine combustor **194** may be independently activated by the control system **400**, according to a turbine engine input control parameter. Such turbine engine input control parameters may include a turbine engine thrust, a temperature of a location in the turbine engine, a temperature of the turbine engine exhaust, a pressure at a location in the turbine engine, a fuel consumption rate, an exhaust moisture content, or a steam consumption rate. This allows for adjusting the amount of combustion and the resultant total thrust from the turbine engine **100** to provide more thrust or less thrust for the particular phase of flight. For example, in a takeoff phase, the control system **400** may supply the fuel from the fuel system **300** and the steam from the steam system **200** to both the primary combustor **26** and the inter-turbine combustor **194** to operate both combustors at or near full capacity in order to generate maximum thrust from the turbine engine **100**. In another example, in a descent phase, the control system may reduce or eliminate the fuel, the steam, or both the fuel and the steam to the primary combustor **26**, the inter-turbine combustor **194**, or both. Reducing or eliminating the fuel reduces total thrust from the turbine engine **100**. In an intermediate phase of flight (e.g., level flight or cruise), the control system may adjust the fuel, the steam, or both the fuel and the steam to the primary combustor **26**, the inter-turbine combustor **194**, or both, to a level between the takeoff levels and the descent levels to accomplish the appropriate amount of thrust to maintain altitude and speed.

(57) The present disclosure, therefore, provides for adjusting the apportionment of steam through the steam manifold **204** to the primary combustor steam supply **206** and the secondary combustor

steam supply **208**, which may allow for manipulating the steam effects on the turbine engine **100**. The present disclosure, therefore, provides for adjusting the apportionment of fuel from the fuel manifold **304** to the primary combustor fuel supply **306** and to the inter-turbine combustor fuel supply **308** and, thus, to the primary combustor **26** and the inter-turbine combustor **194**, respectively. Therefore, the control system **400** controls the apportionment of steam from the steam manifold **204** and fuel from the fuel manifold **304** to the primary combustor **26** and the inter-turbine combustor **194**.

(58) As noted previously, introducing steam into the primary combustor **26** increases the mass flow rate of the primary combustion products **66**, resulting in a net increase in energy introduced into the HPT **28**. Consequently, introducing steam (e.g., from primary combustor steam supply **206**) into the primary combustor **26** increases the net power output by the HPT **28**. Such a net increase in energy, and the resultant net increase in power generated by the HPT **28**, may allow for an overall smaller turbine engine **100** relative to a turbine engine of substantially similar power capacity, or for a larger capacity turbine engine **100** relative to a turbine engine of substantially similar size, without a primary combustor steam supply **206**.

(59) Additionally, introducing steam into the primary combustor **26** may serve to cool the primary combustion products **66**, consequently, cooling elements of the primary combustor **26** and lowering the maximum temperature at or near the exit of the primary combustor **26** (beyond the already lowered temperature that occurs by removing some combustion to the inter-turbine combustor **194**). As previously discussed, lowering the primary combustor **26** maximum temperature may result in lower emissions, improved durability of the turbine engine **100**, or both.

(60) Introducing steam into the inter-turbine combustor **194** increases the mass flow rate of the inter-turbine combustion products **167** flowing through the LPT **30** and lowers the maximum temperature in the inter-turbine combustor **194**. Increasing the mass flow rate of the inter-turbine combustion products **167** results in a net increase in energy introduced into the LPT **30**, and, thus, increases the net power output by the LPT **30**. Such a net increase in energy, and the resultant net increase in power generated by the LPT **30**, may allow for an overall smaller turbine engine **100** relative to a turbine engine of substantially similar power capacity, or for a larger capacity turbine engine **100** relative to a turbine engine of substantially similar size, without the inter-turbine combustor **194** having the inter-turbine combustor steam supply **208**. Lowering the maximum temperature in the inter-turbine combustor **194** further reduces the production of NO_x emissions and reduces the risk of temperature induced degradation of the inter-turbine combustor **194**, the LPT **30**, and other downstream components.

(61) Accordingly, the turbine engine **100**, with both the primary combustor **26** and the inter-turbine combustor **194**, each having steam introduction and fuel introduction, allows independently allocating and apportioning, with the control system **400**, each of the fuel and the steam as desired to optimize various output parameters of the turbine engine **100** according to the operating condition of the turbine engine **100**.

(62) Referring back to the control system **400** of FIG. 2, the fuel manifold control signal **412** provides input data to the fuel manifold **304** to apportion the amount, the rate, or both the amount and the rate of fuel flowing from the fuel supply **302** to the primary combustor fuel supply **306** and the inter-turbine combustor fuel supply **308**. The fuel manifold control signal **412**, therefore, may provide input data to valves, pumps, or other components of the fuel manifold **304** that allow for control of the amount or the rate of flow through the primary combustor fuel supply **306** and the inter-turbine combustor fuel supply **308**. Similarly, the steam manifold control signal **414** provides input data to the steam manifold **204** to apportion the amount, the rate, or both the amount and the rate of steam flowing from the steam supply **202** to the primary combustor steam supply **206** and the inter-turbine combustor steam supply **208**. The steam manifold control signal **414**, therefore, may provide input data to valves, pumps, or other components of the steam manifold **204** that allow for control of the amount or the rate of flow through the primary combustor steam supply

206 and the inter-turbine combustor steam supply **208**. Additionally, supply of the fuel, the steam, or both the fuel and the steam from the fuel manifold **304** and the steam manifold **204**, respectively, may be shut off entirely to one or both of the primary combustor **26** or the inter-turbine combustor **194**, depending on the operating requirements of the turbine engine **100**, as discussed above.

(63) In some examples, the steam supply **202** may be limited or the steam supply **202** may be eliminated. This may be referred to as a limited steam condition, and may occur in situations when less steam is generated in a heat recovery system than may be required for a power setting, in situations with a limited or a finite capacity steam source, or in situations when the steam supply is unavailable. In such conditions of limited steam supply **202**, if temperatures at certain locations of the turbine engine **100** approach a high-temperature threshold (as measured by one or more sensors provided within the turbine engine **100**), the controller **402** may allocate additional steam from the limited steam supply **202** to the position in question, such as the primary combustor **26** or the inter-turbine combustor **194**, to avoid degradation of engine components due to high temperature. Reallocating the steam may reduce overall engine performance, such as total thrust, or may increase emissions, in favor of the higher priority protection the overall safety of the turbine engine **100**. In this example, the limited steam supply **202** condition may result in the controller **402** providing adequate steam from the primary combustor steam supply **206** to keep the maximum temperature of the primary combustor **26** or the HPT **28**, as indicated by the primary combustor data signal **406**, below a predetermined temperature threshold. The controller **402** signals, via the steam manifold control signal **414**, to allocate as a priority over other locations, sufficient steam to the primary combustor **26**, to resolve the high-temperature condition via increased primary combustor steam supply **206**. Upon allocating sufficient steam to the primary combustor **26**, there may be insufficient available steam supply **202** remaining to support the inter-turbine combustor **194**. In such a case, the inter-turbine combustor steam supply **208** will necessarily be reduced due to limited overall steam supply **202**, and the controller **402** may therefore decrease the inter-turbine combustor fuel supply **308** to correlate with the available remaining steam supply **202** for the inter-turbine combustor steam supply **208**, so as not to exceed a high temperature threshold for the inter-turbine combustor **194** or the LPT **30**. A similar control function may occur in the event of a high temperature condition in the inter-turbine combustor **194**.

(64) Additionally or alternatively, the apportionment of the steam supply **202** may be optimized such as to minimize NO_x emissions, to minimize the total fuel consumption, or both. If the steam supply **202** includes extraction of water vapor from the exhaust flow of the turbine engine **100**, the resulting reduction in water vapor in the exhaust may reduce the generation of contrails. Accordingly, increased steam flow may reduce contrail generation.

(65) In some examples, the control system **400** may prioritize providing the steam, the fuel, or both the steam and the fuel to the primary combustor **26** or the inter-turbine combustor **194**, relative to the other. For example, a given quantity of steam introduced at the primary combustor **26** may have a greater net effect on total thrust from the turbine engine **100** than the same quantity of steam introduced at the inter-turbine combustor **194**. In a flight phase such as takeoff, when maximum thrust is desired, and the available steam supply **202** is limited, the controller may prioritize providing steam from the steam supply **202** to the primary combustor **26**, without resulting in blowout in the primary combustor **26**, in order to achieve the greatest thrust benefit, while allocating a relatively lesser amount of steam or allocating the remainder of the available of steam to the inter-turbine combustor **194**.

(66) In some examples, an excess steam condition may be present, wherein the steam supply **202** is greater than the total steam the controller **402** determines to allocate to the primary combustor **26** and the inter-turbine combustor **194**, together, or is greater than the total steam the combustors can accommodate. For example, an idle condition, a descent phase, a cruise phase, precipitation, or overly humid environments can cause an excess steam condition. This occurs when the amount of steam generated exceeds the amount of steam required to operate the primary combustor **26** and the

inter-turbine combustor **194** in the current operating condition. In such circumstances, the controller **402** allocates a predetermined amount of steam to the primary combustor steam supply **206** and the inter-turbine combustor steam supply **208**. The predetermined amount of steam is limited by the amount that each combustor can accommodate safely without resulting in blowout, to avoid loss in total energy or a loss in thrust. In such conditions, when the steam supply **202** is greater than can be accommodated, additional steam is discharged from the turbine engine **100** without passing through the combustors. The above situations are exemplary and the control system **400** may selectively and independently control each of the fuel and the steam to each of the primary combustor **26** and the secondary combustor **194** based on signals received from the combustors (e.g., from sensors or from the primary combustor data signal **406** and the inter-turbine combustor data signal **408**), based on an operating condition of the turbine engine **100**, based on a desired operating condition of the turbine engine **100**, or any combination thereof.

(67) FIG. 3 illustrates a schematic partial view of a turbine engine **500**. The turbine engine **500** is substantially similar to the turbine engine **100** described with respect to FIG. 2, except for the orientation of the secondary combustors, as described below. Accordingly, the same reference numerals will be used for components of the turbine engine **500** that are the same as or similar to the components of the turbine engine **100** and the turbine engine **10** as discussed above. The description of these components above also applies to this embodiment, and a detailed description of these components is omitted here.

(68) The turbine engine **500** includes a fuel system **700**, a steam system **600**, and a control system **800**. The turbine engine **500** includes a primary combustor **26**, a secondary combustor **596**, and a tertiary combustor **598**. The secondary combustor **596** may be a high pressure turbine (HPT) inter-stage combustor **596** and the tertiary combustor **598** may be a low pressure turbine (LPT) inter-stage combustor **598**. Although both the HPT inter-stage combustor **596** and the LPT inter-stage combustor **598** are depicted, only one (either the HPT inter-stage combustor **596** or the LPT inter-stage combustor **598**) may be provided.

(69) In the turbine engine **500**, the HPT **28** and the LPT **30** are each a multi-stage turbine, as described above. The HPT inter-stage combustor **596** is disposed between successive stages of the HPT **28**. That is, the HPT inter-stage combustor **596** is disposed between an HPT first stage **527** of the HPT **28** and an HPT second stage **529** of the HPT **28**, with the HPT first stage **527** and the HPT second stage **529** being adjacent with no intervening stages. The LPT inter-stage combustor **598** is disposed between stages of the LPT **30**. That is, the LPT inter-stage combustor **598** is disposed between an LPT first stage **531** of the LPT **30** and an LPT second stage **533** of the LPT **30**, with the LPT first stage **531** and the LPT second stage **533** being adjacent with no intervening stages.

(70) The steam system **600** includes the steam supply **202**, the steam manifold **204**, and the primary combustor steam supply **206**, described with respect to FIG. 2. The steam system **600** also includes a secondary combustor steam supply **608**, also referred to herein as an HPT inter-stage combustor steam supply **608**, and a tertiary combustor steam supply **610**, also referred to herein as an LPT inter-stage combustor steam supply **610**.

(71) The fuel system **700** includes the fuel supply **302**, the fuel manifold **304**, and the primary combustor fuel supply **306**, described with respect to FIG. 2. The fuel system **700** also includes a secondary combustor fuel supply **708**, also referred to herein as an HPT inter-stage combustor fuel supply **708**, and a tertiary combustor fuel supply **710**, also referred to herein as an LPT inter-stage combustor fuel supply **710**.

(72) The control system **800** is substantially the same as the control system **400**, except that, instead of the inter-turbine combustor data signal **408**, the control system **800** includes a secondary combustor data signal **808** between the controller **402** and the HPT inter-stage combustor **596** and a tertiary combustor data signal **810** between the controller **402** and the LPT inter-stage combustor **598**. The secondary combustor data signal **808**, also referred to herein as an HPT inter-stage combustor data signal **808**, and the tertiary combustor data signal **810**, also referred to herein as an

LPT inter-stage combustor data signal **810**, are substantially the same as the data signals described with respect to FIG. 2, except that the data signals of FIG. 3 provide data from the HPT inter-stage combustor **596** and the LPT inter-stage combustor **598**, respectively.

(73) Operation of the turbine engine **500** is substantially the same as operation of the turbine engine **100** described with respect to FIG. 2.

(74) As discussed above, the primary combustion products **66** flow into the HPT **28**. More specifically, the primary combustion products **66** flow into the HPT first stage **527**, wherein thermal or kinetic energy is extracted from the primary combustion products **66**, contributing to the rotation of the HP shaft **34** and the HPC **24**. Upon exiting the HPT first stage **527**, the primary combustion products **66** enter the HPT inter-stage combustor **596** and are mixed with fuel (from the HPT inter-stage combustor fuel supply **708**) and steam (from the HPT inter-stage combustor steam supply **608**). The mixture is combusted to generate secondary inter-stage combustion products **568**, also referred to herein as HPT inter-stage combustion products **568**, that flow into the HPT second stage **529**. At or near the HPT inter-stage combustor **596**, the HPT inter-stage combustor steam supply **608** introduces steam. The HPT second stage **529** extracts a portion of the thermal energy or the kinetic energy from the HPT inter-stage combustion products **568**, additionally contributing to the rotation of the HP shaft **34** and the HPC **24**. The HPT inter-stage combustion products **568** exit from the HPT **28** and flow into the LPT first stage **531** for further extraction of the thermal energy or the kinetic energy, thereby supporting rotation of the LP shaft **36** and the LPC **22**. Upon exiting the LPT first stage **531**, prior to flowing into the LPT second stage **533**, the HPT inter-stage combustion products **568** flow into the LPT inter-stage combustor **598**. The HPT inter-stage combustion products **568** are mixed with fuel (from the LPT inter-stage combustor fuel supply **710**) and steam (from the LPT inter-stage combustor steam supply **610**) and the mixture is combusted to generate LPT inter-stage combustion products **570**. The LPT inter-stage combustion products **570** then flow through the LPT second stage **533**, where further extraction of the thermal energy or the kinetic energy occurs, additionally supporting rotation of the LP shaft **36** and the LPC **22**.

(75) By conducting a portion of the total combustion in the primary combustor **26**, a portion of the combustion in the HPT inter-stage combustor **596**, and a portion of the combustion in the LPT inter-stage combustor **598**, the total combustion is separated into multiple separate combustion processes, thereby reducing the maximum temperature at the primary combustor **26** relative to a substantially similar capacity turbine engine with a single combustor. The maximum temperature of the HPT inter-stage combustion products **568** occurs at or near the exit of the HPT inter-stage combustor **596** and the maximum temperature of the LPT inter-stage combustion products **570** occurs at or near the exit of the LPT inter-stage combustor **598**. As discussed previously, lowering the temperature of the primary combustion products **66** through the HPT first stage **527** prior to combustion in the HPT inter-stage combustor **596** and lowering the temperature of the HPT inter-stage combustion products **568** prior to combustion in the LPT inter-stage combustor **598**, lowers the overall maximum temperature of the turbine engine **500**, which may occur at or near any of the exit of the primary combustor **26**, the HPT inter-stage combustor **596**, or the LPT inter-stage combustor **598**, relative to a substantially similar capacity turbine engine with a single combustor. The turbine engine **500** thus increases the mass flow rate and reduces the temperatures, as described, with respect to the turbine engine **100**. Likewise, the control system **800** may control the flow of steam and the flow of fuel to each of the HPT inter-stage combustor **596** and the LPT inter-stage combustor **598**, as described, with respect to the control system **400** of the turbine engine **100**.

(76) Although a single inter-stage combustor is illustrated in each of the HPT **28** and the LPT **30**, more or fewer may be provided. For example, two or more inter-stage combustors may be present in the HPT **28** or the LPT **30** or both. In another example, only one of the HPT **28** or the LPT **30** may have one or more inter-stage combustors. Accordingly, any quantity or combination as may be advantageous to the size, shape, layout, manufacture, operation, or performance of the engine may

be provided. Further, a turbine engine with one, three, or any other number of turbines may similarly employ any number of inter-stage combustors. The total number of inter-stage combustors available to each turbine of a turbine engine is equal to one fewer than the number of stages present in the turbine.

(77) FIG. 4 illustrates a schematic partial view of a turbine engine **900**. The turbine engine **900** is substantially similar to the turbine engine **500** described with respect to FIG. 3, except for the manner in which the fuel flow and the steam flow are introduced into each of the combustors, as described below. Accordingly, the same reference numerals will be used for components of the turbine engine **900** that are the same as or similar to the components of the turbine engine **500**, the turbine engine **100**, and the turbine engine **10** as discussed above. The description of these components above also applies to this embodiment, and a detailed description of these components is omitted here. Likewise, the operation of the turbine engine **900** is substantially the same as operation of the turbine engine **500** described with respect to FIG. 3.

(78) FIG. 4 illustrates comingling the fuel flow from the fuel manifold **304** and the steam flow from the steam manifold **204** prior to entry into the combustors. For example, the primary combustor fuel supply **306** and the primary combustor steam supply **206** are combined in a primary combustor fuel and steam supply **906** and introduced into the primary combustor **26** as a combined fuel-steam flow. The HPT inter-stage combustor fuel supply **708** and the HPT inter-stage combustor steam supply **608** are combined in an HPT inter-stage combustor fuel and steam supply **908** (also referred to herein as a secondary inter-stage combustor fuel and steam supply **908**) and introduced into the HPT inter-stage combustor **596**. The LPT inter-stage combustor fuel supply **710** and the LPT inter-stage combustor steam supply **610** are combined in an LPT inter-stage combustor fuel and steam supply **910** (also referred to herein as a tertiary inter-stage combustor fuel and steam supply **910**) and introduced into the LPT inter-stage combustor **598**.

(79) Joining the fuel and steam supplies as described may enhance mixing of the fuel and the steam prior to introduction into the primary combustor **26**, the HPT inter-stage combustor **596**, and the LPT inter-stage combustor **598**, respectively. Enhanced mixing may allow for more even fuel burn, may prevent partial blowout, or both. Additionally or alternatively, such joining may leverage the limited space within the turbine engine **900**.

(80) Although illustrated for each of the combustors, joining the fuel and the steam may be performed in part or in whole. In other words, one or more of the fuel and steam supply for the primary combustor **26**, the HPT inter-stage combustor **596**, or the LPT inter-stage combustor **598** may have a joined a fuel and steam supply, while one or more may not.

(81) The examples illustrated in FIGS. 2 to 4 are exemplary and various features therein may be combined among the examples. For example, although shown in the example of FIG. 3, the fuel supply and the steam supply of the turbine engine **100** (FIG. 2) may optionally be joined on the primary combustor fuel supply **306** and the primary combustor steam supply **206** for the primary combustor **26**, on the inter-turbine combustor fuel supply **308** and the inter-turbine combustor steam supply **208**, or both.

(82) FIG. 5 illustrates a multi-stage turbine **1000** that may be either or both of the HPT **28** or the LPT **30** having an inter-stage combustor as described with respect to FIG. 4. Either or both of the HPT inter-stage combustor **596** or the LPT inter-stage combustor **598** described with respect to FIG. 4 may include multiple distinct inter-stage combustors between multiple stages of the turbine, as described in more detail to follow. The multi-stage turbine **1000** contains rotor blades **1028** and stator vanes **1030** corresponding to the stages of the multi-stage turbine **1000**. The exemplary multi-stage turbine **1000** of FIG. 5 has four-stages. More or fewer may, however, be provided. The multi-stage turbine **1000** may include one or more inter-stage combustors, such as a first inter-stage combustor **1012**, a second inter-stage combustor **1014**, and a third inter-stage combustor **1016**. Although three interstage combustors are illustrated and described, more or fewer may be provided. For example, a turbine of n stages may contain up to $n-1$ inter-stage combustors. Although

illustrated among the first stage, the second stage, and the third stage of the multi-stage turbine **1000**, one or more of the inter-stage combustors may be omitted, or the inter-stage combustors may begin at a second stage or a third stage of the multi-stage turbine **1000**, etc., or both.

(83) The first inter-stage combustor **1012**, the second inter-stage combustor **1014**, and the third inter-stage combustor **1016**, may be disposed among successive rotor blades of the successive stages, successive stator vanes of the successive stages, or between a rotor blade of one stage of the successive stages and a stator vane of the other stage of the successive stages.

(84) The first inter-stage combustor **1012** receives first combustion products **1066** from upstream of the first inter-stage combustor **1012**. In examples with the multi-stage turbine **1000** following the primary combustor **26** (FIG. 3), the first combustion products **1066** are the primary combustion products **66** (FIG. 3). In examples with the multi-stage turbine **1000** following another turbine, such as in examples when the multi-stage turbine **1000** is the LPT **30**, the first combustion products **1066** are the secondary inter-stage combustion products **568** (FIG. 3). The first inter-stage combustor **1012** receives the first combustion products **1066**, a first inter-stage combustor fuel supply **1006** from a fuel manifold **1004**, and a first inter-stage combustor steam supply **1022** from a steam manifold **1020**, which are mixed and combusted in the first inter-stage combustor **1012**, generating first inter-stage combustion products **1068**. The second inter-stage combustor **1014** receives the first inter-stage combustion products **1068** from the first inter-stage combustor **1012**, a second inter-stage combustor fuel supply **1008** from the fuel manifold **1004**, and a second inter-stage combustor steam supply **1024** from the steam manifold **1020**, which are mixed and combusted in the second inter-stage combustor **1014**, generating second inter-stage combustion products **1070**.

(85) The third inter-stage combustor **1016** receives the second inter-stage combustion products **1070** from the second inter-stage combustor **1014**, a third inter-stage combustor fuel supply **1010** from the fuel manifold **1004**, and a third inter-stage combustor steam supply **1026** from the steam manifold **1020**, which is then combusted in the third inter-stage combustor **1016**, generating third stage inter-stage combustion products **1072**.

(86) The fuel manifold **1004** and the steam manifold **1020** may be included in respective fuel systems and steam systems, such as described previously. Accordingly, the fuel manifold **1004** may provide fuel from a fuel supply **1002** and the steam manifold **1020** may provide steam from a steam supply **1018**.

(87) Operation of the multi-stage turbine **1000** is substantially the same as described with respect to the turbines of FIGS. 1 to 4. In the multi-stage turbine **1000** having multiple inter-stage combustors, the inter-stage combustors may be in successive stages or non-successive stages of the turbine. In other words, a three-stage turbine may employ inter-stage combustors in stages one, two, and three, or may employ combustors at stages one and three, or may employ inter-stage combustors and an inter-stage combustor steam supply in any other combination as may be advantageous to the size, shape, layout, manufacture, operation, or performance of the engine. This is merely exemplary and other arrangements of inter-stage combustors are contemplated. In some examples, the multi-stage turbine **1000** may employ multiple inter-stage combustors, but may only employ a steam supply for a portion of the inter-stage combustors.

(88) In some examples, a turbine engine may employ both an inter-turbine combustor (e.g., inter-turbine combustor **194** of FIG. 2) and one or more inter-stage combustors (e.g., inter-stage combustors described in FIGS. 3 to 5), as described herein, with steam supply to one or more of the inter-turbine or inter-stage combustors.

(89) FIGS. 6 and 7 illustrate exemplary combustors that may be employed as the primary combustor, one or more of the inter-turbine combustors, one or more of the inter-stage combustors, or combinations thereof, as described with respect to FIGS. 1 to 5.

(90) For example, FIG. 6 illustrates a tangential-radial inflow (TRI) combustor **1100**. The TRI combustor **1100** may be a primary combustor, an inter-turbine combustor, or an inter-stage

combustor. The TRI combustor **1100** generally defines an axial direction A extending along an axial centerline **1102**, a radial direction R, and a circumferential direction C (i.e., a direction extending about the axial direction A). The axial centerline **1102** may align with a centerline of the turbine engine within which the TRI combustor **1100** is installed (e.g., the longitudinal centerline axis **12** of the turbine engine **10** of FIG. 1).

(91) The TRI combustor **1100** includes an inner combustion liner **1104** and an outer combustion liner **1106**. The inner combustion liner **1104** and the outer combustion liner **1106** define a combustion chamber **1108** having a combustor outlet **1110**. The TRI combustor **1100** includes an inlet assembly **1112** that introduces an airflow **1114**, such as compressed air, from a compressor section of the turbine engine (in the case of a primary combustor) or an upstream turbine or an upstream turbine stage (in the case of an inter-turbine combustor or an inter-stage combustor) into the combustion chamber **1108**. The inlet assembly **1112** introduces the airflow **1114** in a manner such that the airflow **1114** has a desired swirl.

(92) In addition, the inner combustion liner **1104** includes a plurality of dilution holes **1116** to provide a dilution airflow **1118** to the combustion chamber **1108**. The exemplary dilution holes **1116** are configured such that dilution airflow **1118** discharged therefrom flows helically relative to the axial centerline **1102** of the TRI combustor **1100**, such that an angular momentum of the airflow **1114** is maintained when dilution airflow **1118** mixes with the airflow **1114**. Each dilution hole **1116** may include a chute **1120** to facilitate channeling airflow from a source (not shown) through dilution holes **1116**.

(93) As described above, an embodiment of the primary combustor, the inter-turbine combustor, or the inter-stage combustor described herein relates to a TRI combustor **1100**. In the TRI combustor **1100**, the inner combustion liner **1104** and the outer combustion liner **1106** are convex relative to the axial centerline **1102** of the TRI combustor **1100** such that combustion chamber **1108** is defined at a radially outermost region of the TRI combustor **1100**. To facilitate inducing bulk swirl in the airflow **1114**, the inlet assemblies **1112** are oriented to discharge the airflow **1114** circumferentially and radially into combustion chamber **1108**.

(94) One or more of the inlet assemblies **1112** may introduce steam from the steam supply, fuel from the fuel supply, or both, as previously described herein. Additionally or alternatively, the steam supply or the fuel supply may be introduced at other locations along the length of the TRI combustor **1100**, upstream of the TRI combustor **1100**, downstream of the TRI combustor **1100**, or any combination thereof.

(95) FIG. 7 illustrates schematic view of an exemplary trapped vortex (TV) combustor **1200** that may be used in a gas turbine engine. For example, the exemplary TV combustor **1200** of FIG. 7 may be configured in a similar manner as any combustor, any combination of the combustors, or all combustors described here, such as, the primary combustor **26** incorporated in the turbine engine **10** of FIG. 1, the primary combustor **26** or the inter-turbine combustor **194** incorporated in the turbine engine **100** of FIG. 2, the primary combustor **26**, the HPT inter-stage combustor **596**, or the LPT inter-stage combustor **598** incorporated in the turbine engine **500** of FIGS. 3 and 4, or the first inter-stage combustor **1012**, the second inter-stage combustor **1014**, or the third inter-stage combustor **1016** incorporated in the multi-stage turbine **1000** of FIG. 5.

(96) FIG. 7 shows the TV combustor **1200**, and the combustor may be included within a gas turbine engine disclosed herein. The TV combustor **1200** comprises a TV combustion zone **1212**, and a combustor exit **1218**.

(97) In order to simplify the illustration and the description, only the upper half portion of the TV combustor **1200** in FIG. 7 is indicated by reference numbers and described specifically. Accordingly, the opposite lower half portion could be understood totally by reference to the illustration and the description of the upper half portion, since the TV combustor **1200** is substantially symmetrical about a longitudinal centerline axis **12** of the gas turbine.

(98) Alternatively, the TV combustor as employed in an inter-turbine combustor or an inter-stage

combustor, may be single sided. That is, the combustor may not be fully annular, but may instead be an annular section.

(99) The TV combustor **1200** comprises an annular combustor that is shaped as generally annular about the longitudinal centerline axis **12** of the gas turbine, such that the TV combustion zone **1212** may be shaped as annular. The TV combustion zone **1212** may be formed or shaped as a trapped vortex (TV) combustion cavity in various embodiments. A combustor casing (not shown) may be positioned around the combustor for providing support or protection, and the like.

(100) As illustrated in FIG. 7, the upper half portion and the lower half portion of the TV combustion zone **1212** each comprises a side wall **1214**, at least one pilot fuel nozzle **1220** disposed on one side end (forward end) of the side wall **1214**, and an igniter **1222** disposed on a radially outward end of the side wall **1214** for igniting. A plurality of pilot fuel nozzles **1220** may be disposed symmetrically about the longitudinal centerline axis **12**, such as being disposed circumferentially surrounding the longitudinal centerline axis **12**.

(101) As described above, the TV combustion zone **1212** may have a substantially circular longitudinal cross-sectional shape depicted in FIGS. 1 to 3. In other exemplary embodiments, the TV combustion zone may be configured as substantially arcuate in longitudinal cross section or substantially rectangular in longitudinal cross section.

(102) The one or more pilot fuel nozzles **1220** are operable to inject a fuel (or reactant) into the TV combustion zone **1212**. The pilot fuel nozzle(s) **1220** may be air-blast nozzle(s), pressure atomizer nozzle(s), plain jet orifice nozzle(s), or any other kinds of nozzles that one skilled in the art could conceive. The fuel comprises a liquid fuel, a gaseous fuel, or a combination of these, which can be selected from the usual fuels, such as jet fuel and any other kinds of fuel that any person skilled in the art could conceive. Air **1224** is compressed air from a compressor (not shown) disposed upstream of the TV combustor **1200**, and the air **1224** is directed into the TV combustion zone **1212** via a plurality of air apertures (not shown) formed through the side wall **1214** along a periphery of the TV combustion zone **1212** and flows toroidally and enhances the mixing effect with the fuel.

(103) The fuel and the air **1224** are received and mixed in the TV combustion zone **1212**. The igniter **1222** initiates combustion by a spark to produce combustion products P flowing toroidally therein.

(104) The steam supply may be introduced into the TV combustor **1200** through one or more steam inlets **1226**. Additionally or alternatively, the steam supply may be combined with air and fuel upstream or downstream of the TV combustor **1200**.

(105) Although not shown, further embodiments of the TV combustor **1200** may include a secondary, tertiary, or more combustion zones disposed downstream of the TV combustion zone **1212**.

(106) A turbine engine with multiple combustors such as primary, inter-turbine, and inter-stage combustors may employ TRI combustors, TV combustors, or a combination of TRI combustors and TV combustors in any combination as may be advantageous to the size, shape, layout, manufacture, operation, or performance of the engine. Further, the turbine engine may include combustor types not herein contemplated, as may be so advantageous.

(107) Introducing steam into various locations of a turbine engine as described herein, may occur at locations within, adjacent to, or separate from the high-temperature locations of the turbine engine, as may be advantageous to the size, the shape, the layout, the manufacture, the operation, or the performance of the engine. For example, the steam may be introduced to a core air flow path upstream or downstream of a combustor, or the steam may be injected through passages in the stator vanes of a turbine. When so introduced, at locations adjacent to or separate from the high-temperature location, the steam injection may raise the temperature at the point of injection, while still serving to lower the temperature in a more critical high-temperature location of the turbine engine.

(108) Turbine engines see higher maximum internal temperatures as a consequence of designs intended to reduce fuel consumption and CO₂ emission. The higher internal temperatures result in the emission of other pollutants, including nitrogen oxides emissions (NO_x emissions), known to cause damage to the atmosphere, and believed to contribute to global warming and/or climate change. By introducing steam in conjunction with secondary inter-turbine or inter-stage combustors (ITB's) in a turbine engine with high internal pressures and temperatures, harmful emissions may be reduced, and thrust may be increased relative to a similar capacity turbine engine with a single combustor, or without steam at the one or more ITB's.

(109) The foregoing description describes a fuel-lean burn condition, where all fuel is consumed at the respective combustors, leaving unconsumed air in combustion products to be mixed with fuel and further combusted downstream. Further embodiments are contemplated with a similar function, but with a fuel-rich burn condition. In a fuel-rich burn condition, the foregoing fuel systems as described are substituted with analogous air systems.

(110) Further aspects of the present disclosure are provided by the subject matter of the following clauses.

(111) A turbine engine comprises a compressor section that provides a compressed air flow, a fuel system that provides a fuel to a primary combustor fuel supply and a secondary combustor fuel supply, a primary combustor located downstream of the compressor section, the primary combustor receiving the compressed air flow and the primary combustor fuel supply, and combusting the compressed air flow and the primary combustor fuel supply to generate primary combustion products, a turbine section located downstream of the primary combustor, the turbine section comprises a turbine that is caused to rotate by the primary combustion products, and a secondary combustor that combusts a mixture of the primary combustion products and the secondary combustor fuel supply to generate secondary combustion products, and a steam system comprises a steam supply to the secondary combustor to increase total mass flow in the turbine section.

(112) The turbine engine of the preceding clause, wherein the turbine is a first turbine, the turbine section further comprises a second turbine, wherein the secondary combustor is an inter-turbine combustor located between the first turbine and the second turbine, the secondary combustor fuel supply is an inter-turbine combustor fuel supply, the steam supply is an inter-turbine combustor steam supply, and the secondary combustion products are inter-turbine combustion products, and the inter-turbine combustion products and the inter-turbine combustor steam supply rotate the second turbine.

(113) The turbine engine of any preceding clause, wherein the steam supply is a secondary combustor steam supply and the steam system further comprises a primary combustor steam supply that provides steam to the primary combustor.

(114) The turbine engine of the preceding clause, wherein the fuel system further comprises a fuel manifold that apportions a flow of fuel to the primary combustor fuel supply and to the secondary combustor fuel supply, the steam system further comprises a steam manifold that apportions a flow of steam to the primary combustor steam supply and to the secondary combustor steam supply, and the turbine engine further comprises a controller that receives a primary combustor data signal indicating a state of the primary combustor and a secondary combustor data signal indicating a state of the secondary combustor, processes the primary combustor data signal and the secondary combustor data signal, and generates a fuel manifold control signal that controls the fuel manifold and a steam manifold control signal that controls the steam manifold.

(115) The turbine engine of the preceding clause, wherein the primary combustor data signal or the secondary combustor data signal indicates at least one of a temperature, a pressure, or a mass flow rate.

(116) The turbine engine of any preceding clause, wherein the secondary combustor fuel supply is an inter-stage combustor fuel supply, the steam supply is an inter-stage combustor steam supply, the turbine is a multi-stage turbine, and the secondary combustor is an inter-stage combustor located

between adjacent stages such that the secondary combustion products are inter-stage combustion products, and the inter-stage combustion products and the inter-stage combustor steam supply rotate the multi-stage turbine.

(117) The turbine engine of the preceding clause, wherein the multi-stage turbine is a first multi-stage turbine, and the inter-stage combustion products are first inter-stage combustion products, and the turbine section further comprises a second multi-stage turbine that includes a second inter-stage combustor between adjacent stages of the second multi-stage turbine that combusts the first inter-stage combustion products and a second inter-stage combustor fuel supply, to produce second inter-stage combustion products.

(118) The turbine engine of any preceding clause, wherein the inter-stage combustor is a first inter-stage combustor, the inter-stage combustor fuel supply is a first inter-stage combustor fuel supply, the inter-stage combustor steam supply is a first inter-stage combustor steam supply, and the inter-stage combustion products are first inter-stage combustion products, the multi-stage turbine comprising a second inter-stage combustor between adjacent stages, the fuel system comprising a second inter-stage combustor fuel supply providing fuel to the second inter-stage combustor, and the steam system comprising a second inter-stage combustor steam supply providing steam to the second inter-stage combustor to increase total mass flow in the turbine section, wherein the second inter-stage combustor further combusts the first inter-stage combustion products and the second inter-stage combustor fuel supply, generating second inter-stage combustion products.

(119) The turbine engine of any preceding clause, wherein the turbine section comprises a second turbine, and an inter-turbine combustor located between the multi-stage turbine and the second turbine, wherein the fuel system comprises an inter-turbine combustor fuel supply providing fuel to the inter-turbine combustor, the steam system comprises an inter-turbine combustor steam supply that provides steam to the inter-turbine combustor, and the inter-turbine combustor further combusts the inter-stage combustion products and the inter-turbine combustor fuel supply to produce inter-turbine combustion products.

(120) A method of operating the turbine engine of any preceding clause, the method comprising apportioning a fuel flow to the primary combustor fuel supply, apportioning a fuel flow to the secondary combustor fuel supply, and apportioning a steam flow to the steam supply, wherein the apportioning the fuel flow and the apportioning the steam flow are based on at least one of a turbine engine input control parameter, a primary combustor data signal, or a secondary combustor data signal.

(121) The method of the preceding clause, wherein the primary combustor data signal or the secondary combustor data signal indicates at least one of a temperature, a pressure, or a mass flow rate.

(122) The method of any preceding clause, wherein, in an excess steam condition, the method discharges steam from the turbine engine.

(123) The method of any preceding clause, wherein the steam supply is a secondary combustor steam supply, the steam system further comprises a primary combustor steam supply providing steam to the primary combustor, and the method further comprises apportioning the steam flow to the primary combustor steam supply.

(124) The method of the preceding clause, wherein during a high power condition, the method increases the steam flow to the primary combustor steam supply, the steam flow to the secondary combustor steam supply, or both the steam flow to the primary combustor steam supply and the steam flow to the secondary combustor steam supply.

(125) The method of any preceding clause, further comprising apportioning the primary combustor steam supply, the secondary combustor steam supply, the primary combustor fuel supply, and the secondary combustor fuel supply to reduce fuel consumption for a given turbine engine input control parameter.

(126) The method of any preceding clause, further comprising controlling the primary combustor

steam supply or the secondary combustor steam supply to prevent blowout.

(127) The method of any preceding clause, further comprising apportioning the primary combustor steam supply, the secondary combustor steam supply, the primary combustor fuel supply, and the secondary combustor fuel supply to reduce contrail generation.

(128) The method of any preceding clause, further comprising apportioning the primary combustor steam supply, the secondary combustor steam supply, the primary combustor fuel supply, and the secondary combustor fuel supply to reduce NO_x emission.

(129) The method of any preceding clause, wherein, upon the primary combustor data signal or the secondary combustor data signal reaching a predetermined threshold value, the method further comprises increasing the primary combustor steam supply, the secondary combustor steam supply, or both the primary combustor steam supply and the secondary combustor steam supply.

(130) The method of the preceding clause, wherein, in a limited steam condition, the method further comprises decreasing the primary combustor fuel supply, the secondary combustor fuel supply, or both the primary combustor fuel supply and the secondary combustor fuel supply.

(131) The turbine engine of any preceding clause, further comprising a core turbine engine casing containing a core air supply, wherein the secondary combustor steam supply is fluidly coupled to the core turbine engine casing, introducing steam to the core air, upstream of the secondary combustor.

(132) The turbine engine of any preceding clause, wherein the secondary combustor steam supply is fluidly coupled to the secondary combustor.

(133) The turbine engine of any preceding clause, wherein the secondary combustor steam supply and the secondary combustor fuel supply are fluidly coupled to a secondary combustor fuel and steam supply, which is fluidly coupled to the secondary combustor.

(134) The turbine engine of any preceding clause, the turbine section further comprising at least one stator vane, and at least one passage in the at least one stator vane, wherein the secondary combustor steam supply is introduced through the at least one passage in the at least one stator vane.

(135) The turbine engine of any preceding clause, wherein at least one of the primary combustor or the secondary combustor is a tangential-radial inflow combustor.

(136) The turbine engine of any preceding clause, wherein at least one of the primary combustor or secondary combustor is a trapped vortex combustor.

(137) A method of operating the turbine engine of any preceding clause, the method comprising apportioning the primary combustor fuel supply and the secondary combustor fuel supply and apportioning the primary combustor steam supply and the secondary combustor steam supply, based on a turbine engine input control parameter, a primary combustor data signal, and a secondary combustor data signal.

(138) The turbine engine of any preceding clause, wherein the fuel system further comprises a fuel manifold that apportions a flow of fuel to the primary combustor fuel supply and to the secondary combustor fuel supply, the steam system further comprises a steam manifold that apportions the flow of steam to the secondary combustor, and the turbine engine further comprises a control system generating a primary combustor data signal that communicates a signal of a state of the primary combustor, a secondary combustor data signal that communicates a signal of a state of the secondary combustor, a fuel manifold control signal that controls the fuel manifold, a steam manifold control signal that controls the steam manifold, and a controller that determines and generates the fuel manifold control signal and the steam manifold control signal.

(139) Although the foregoing description is directed to the preferred embodiments of the present disclosure, other variations and modifications will be apparent to those skilled in the art and may be made without departing from the spirit or the scope of the disclosure. Moreover, features described in connection with one embodiment of the present disclosure may be used in conjunction with other embodiments, even if not explicitly stated above.

Claims

1. A turbine engine comprising: a compressor section including an HP compressor and an LP compressor, the compressor section providing a compressed air flow; a fuel system that provides fuel to a primary combustor fuel supply and a secondary combustor fuel supply; a primary combustor located downstream of the compressor section, the primary combustor receiving the compressed air flow and the primary combustor fuel supply, and combusting the compressed air flow and the primary combustor fuel supply to generate primary combustion products; a turbine section located downstream of the primary combustor, the turbine section comprising: an HP turbine mechanically connected to the HP compressor via an HP shaft, the HP turbine rotating from a flow of the primary combustion products; an LP turbine mechanically connected to the LP compressor via an LP shaft, the LP turbine positioned downstream of the HP turbine; and a secondary combustor, positioned between the HP turbine and the LP turbine, that combusts a mixture of the primary combustion products and the secondary combustor fuel supply to generate secondary combustion products; and a steam system comprising a steam supply to supply steam to the secondary combustor to increase total mass flow in the turbine section.
2. The turbine engine of claim 1, wherein the steam supply is a secondary combustor steam supply and the steam system further comprises a primary combustor steam supply that provides steam to the primary combustor.
3. The turbine engine of claim 2, wherein the fuel system further comprises a fuel manifold that apportions a flow of fuel to the primary combustor fuel supply and to the secondary combustor fuel supply, the steam system further comprises a steam manifold that apportions a flow of steam to the primary combustor steam supply and to the secondary combustor steam supply, and the turbine engine further comprises a controller that receives a primary combustor data signal indicating a state of the primary combustor and a secondary combustor data signal indicating a state of the secondary combustor, processes the primary combustor data signal and the secondary combustor data signal, and generates a fuel manifold control signal that controls the fuel manifold and a steam manifold control signal that controls the steam manifold.
4. The turbine engine of claim 3, wherein the primary combustor data signal or the secondary combustor data signal indicates at least one of a temperature, a pressure, or a mass flow rate.
5. A method of operating the turbine engine of claim 1, the method comprising: apportioning a fuel flow to the primary combustor fuel supply; apportioning a fuel flow to the secondary combustor fuel supply; and apportioning a steam flow to the steam supply, wherein the apportioning the fuel flow and the apportioning the steam flow are based on at least one of a turbine engine input control parameter, a primary combustor data signal, or a secondary combustor data signal.
6. The method of claim 5, wherein the primary combustor data signal or the secondary combustor data signal indicates at least one of a temperature, a pressure, or a mass flow rate.
7. The method of claim 5, wherein, in an excess steam condition, the method discharges steam from the turbine engine.
8. The method of claim 5, wherein the steam supply is a secondary combustor steam supply, the steam system further comprises a primary combustor steam supply providing steam to the primary combustor, and the method further comprises apportioning the steam flow to the primary combustor steam supply.
9. The method of claim 8, wherein, during a high power condition, the method increases the steam flow to the primary combustor steam supply, the steam flow to the secondary combustor steam supply, or both.
10. The method of claim 8, further comprising apportioning the primary combustor steam supply, the secondary combustor steam supply, the primary combustor fuel supply, and the secondary combustor fuel supply to reduce fuel consumption for a given turbine engine input control

parameter.

11. The method of claim 8, further comprising controlling the primary combustor steam supply or the secondary combustor steam supply to prevent blowout.

12. The method of claim 8, further comprising apportioning the primary combustor steam supply, the secondary combustor steam supply, the primary combustor fuel supply, and the secondary combustor fuel supply to reduce contrail generation.

13. The method of claim 8, further comprising apportioning the primary combustor steam supply, the secondary combustor steam supply, the primary combustor fuel supply, and the secondary combustor fuel supply to reduce NO.sub.x emission.

14. The method of claim 8, wherein, upon the primary combustor data signal or the secondary combustor data signal reaching a predetermined threshold value, the method further comprises increasing the primary combustor steam supply or the secondary combustor steam supply.

15. The method of claim 14, wherein, in a limited steam condition, the method further comprises decreasing the primary combustor fuel supply or the secondary combustor fuel supply.

16. A turbine engine comprising: a compressor section that provides a compressed air flow; a fuel system that provides fuel to a primary combustor fuel supply and an inter-stage combustor fuel supply; a primary combustor located downstream of the compressor section, the primary combustor receiving the compressed air flow and the primary combustor fuel supply, and combusting the compressed air flow and the primary combustor fuel supply to generate primary combustion products; a turbine section located downstream of the primary combustor, the turbine section comprising: an HP turbine connected to an HP shaft and rotating from a flow of the primary combustion products, the HP turbine being a multi-stage turbine; an LP turbine connected to an LP shaft and positioned downstream of the HP turbine; and a secondary combustor that combusts a mixture of the primary combustion products and fuel from the inter-stage combustor fuel supply to generate inter-stage combustion products, the secondary combustor being an inter-stage combustor located between adjacent stages of the HP turbine such that the inter-stage combustion products of the secondary combustor are HP inter-stage combustion products; and a steam system comprising an inter-stage combustor steam supply to supply steam to the secondary combustor to increase total mass flow in the turbine section, wherein the HP inter-stage combustion products and the inter-stage combustor steam supply rotate the HP turbine.

17. The turbine engine of claim 16, wherein the inter-stage combustor fuel supply includes an HP inter-stage combustor fuel supply and an LP inter-stage combustor fuel supply, and the LP turbine includes adjacent stages and a tertiary combustor between the adjacent stages, the tertiary combustor combusting the HP inter-stage combustion products and the fuel from the LP inter-stage combustor fuel supply, to produce LP inter-stage combustion products.

18. The turbine engine of claim 16, further comprising an inter-turbine combustor fuel supply, wherein the turbine section further comprises an inter-turbine combustor positioned between the HP turbine and the LP turbine and receiving fuel from the inter-turbine combustor fuel supply, and wherein the steam system further comprises an inter-turbine combustor steam supply that provides steam to the inter-turbine combustor, and the inter-turbine combustor further combusts the HP inter-stage combustion products and the inter-turbine combustor fuel supply to produce inter-turbine combustion products.

19. A method of operating the turbine engine of claim 16, the method comprising: apportioning a fuel flow to the primary combustor fuel supply; apportioning a fuel flow to the inter-stage combustor fuel supply; and apportioning a steam flow to the inter-stage combustor steam supply, wherein the apportioning the fuel flow and the apportioning the steam flow are based on at least one of a turbine engine input control parameter, a primary combustor data signal, or a secondary combustor data signal.

20. The method of claim 19, wherein the primary combustor data signal or the secondary combustor data signal indicates at least one of a temperature, a pressure, or a mass flow rate.

